

Java programming exercise that focuses on:

- **Array of objects**
- **Passing arrays to methods**
- **Static methods**
- **Basic OOP structure**

Included:

- The **structured question with blanks** (for students to complete)
- A **sample output**
- The **complete Java solution**

Structured Question: Java – Arrays of Objects & Methods

Case Study: Staff Salary Management

You are required to write a Java program that models a simple **Staff Salary System**.

Requirements:

1. Define a class called Staff with the following:
 - A **String** field for name
 - A **double** field for salary
 - A **constructor** to initialize both fields
 - A method to **display** the staff details
2. In the Main class:
 - Create an array of **3 Staff objects**
 - Pass this array to a **static method** named printAllStaff() that:
 - Displays details of all staff
 - Pass the array to another static method findHighestSalary() that:
 - Finds and displays the staff member with the **highest salary**

Fill in the Blanks: (Student task to complete the blank space)

```
class Staff {  
    String name;  
    double salary;
```


// Constructor

```
public Staff(String name, double salary) {
```

```
    this.name = name;
```

```
    this.salary = salary;
```

```
}
```

```
public void display() {
```

```
    System.out.println("Name: " + name + ", Salary: " + salary);
```

```
}
```

```
}
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        // Create array of Staff objects
```

```
        Staff[] staffList = new Staff[3];
```

```
        staffList[0] = new Staff("Zaid", 50000);
```

```
        staffList[1] = new Staff("Ben", 62000);
```

```
        staffList[2] = new Staff("Sarah", 58000);
```

```
        // Call method to print all staff
```

```
        printAllStaff(staffList);
```

```
        // Call method to find highest salary
```

```
        findHighestSalary(staffList);
```

```
}
```

```
// Method to print all staff
```

```
public static void printAllStaff(Staff[] staffList) {
```

```
    for (Staff s : staffList) {
```

```
        s.display();
```



```
    }  
}  
  
// Method to find highest salary  
public static void findHighestSalary(Staff [] staffList) {  
    Staff highest = staffList[0];  
    for (int i = 1; i < staffList.length; i++) {  
        if (staffList[i].salary > highest.salary) {  
            highest = staffList[i];  
        }  
    }  
    System.out.println("Staff with Highest Salary:");  
    highest.display();  
}  
}
```

Sample Output

```
Name: Zaid, Salary: 50000.0  
Name: Ben, Salary: 62000.0  
Name: Sarah, Salary: 58000.0  
Staff with Highest Salary:  
Name: Ben, Salary: 62000.0
```

```
    }  
}  
  
// Method to find highest salary  
public static void findHighestSalary(_____) {  
    Staff highest = staffList[0];  
    for (int i = 1; i < staffList.length; i++) {  
        if (staffList[i].salary > _____.salary) {  
            highest = staffList[i];  
        }  
    }  
    System.out.println("Staff with Highest Salary:");  
    highest.display();  
}  
}
```

Sample Output

```
Name: Zaid, Salary: 50000.0  
Name: Ben, Salary: 62000.0  
Name: Sarah, Salary: 58000.0  
Staff with Highest Salary:  
Name: Ben, Salary: 62000.0
```
